Binary Batch Distillation

Pre-Lab Plan

Unit Operations Lab #2

09/17/2025

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1. Experimental Objective

The objective of this experiment is to determine the boil-up rate, pressure drop, and overall distillation column efficiency of a separation process of a binary ethanol-water mixture. This is done by finding the tray temperatures, compositions, and pressure drop and calculating key design and performance parameters of the distillation process.

2. Experimental

The purpose of this experiment is to investigate binary batch distillation of an ethanol-water using an Armfield UOP3CC distillation column. The study focuses on evaluating how operating conditions such as the boil-up rate and reflux ratio influence the performances, such as pressure drop, product composition, and overall column efficiency. By collecting temperature, pressure, and composition data under controlled conditions, such as the reflux ratio and the power intake. The experiment links measured variables to theoretical predictions from the McCabe-Thiele method, which allows comparison between ideal and actual column behavior.

a. Apparatus

Table 1. Apparatus for Evaporation Experiment		
No.	**Component**	**Description**
1	Condenser	Cools and liquefies the product
2	Decanter	Collects condensed product and allows part of it to be returned to the column
3	Reboiler	Heats the solution to generate vapor
4	Distillation Column	Provides trays for repeated vapor-liquid contact and separation of ethanol and water
5	Bottom Receiver	Collects water (heavier component) leaving the column

6	Top Receiver	Collects the distillated ethanol
7	Manometer	Measures the pressure of the column
8	Cooling Water Thermometer	Monitors temperature of cooling water
9	Thermocouples	They measure the temperatures at different parts of the distillation column

b. Materials and Supplies:

Table 2. **Materials and Supplies for Evaporation Experiment**	
Materials & Supplies	**Description**
Ethanol	200 proof Ethanol (100% abv).
DI Water	Water used to make ethanol & water mixture.
Large Beaker	Used to 10 L solution.
Snap 41	Reads alcohol level of sample collected.
Graduated Cylinder	Used to measure water and ethanol volume levels.
Collecting flasks	Used to measure collected bottoms product.
Control Console	Used to control power supply to the reboiler, and control reflux ratio.
Funnel	Used to pour ethanol water solution into the reboiler.
Distillation Column	Binary Distillation column to separate alcohol water solution based on volatility.

c. Experimental Plan

Overview

Feed Preparation & Setup

- Prepare 10L of 10wt%ethanol in deionized water by mass
- Add the feed mixture to the reboiler using a funnel; replace the filler cap firmly.
- Introduction of condenser cooling water with a rate of approximately 3 L/min.
- Before heating, ensure that all the valves are in place.
- **In experiment A**, the total reflux run was performed.
- Turn on the reboiler heater (0.751.0 kW) to start boiling.
- Place the reflux divider to a total reflux to keep all the condensate in the column.
- Wait for 15 minutes until equilibrium (steady temperatures and tray bubbling).
- Measure temperature of the top and bottom columns at regular intervals of 2 minutes until the temperature levels off.
- Determine the rate of boil-up by assembling condensate in a graduated cylinder and timing collection (3 times, average results).
- Take overheads and bottoms samples (dispose of initial 5-10 mL and collect from thereof about 10 mL to be analysed).

Experiment B Finite Reflux Ratio Runs.

- Modulate reflux controller to a given ratio (e.g., 5:1).
- Wait up to 10 minutes to stabilize the column.
- Record highest and lowest records set.
- Determine boil-up rate 3 times.

- Take a sample of overhead and bottoms after every 5 minutes (at least 3 sets).
- Repeat with a different reflux ratio (e.g. 3:1).

Shutdown & Cleanup

- Turn off the reboiler heater and cool down the system.
- Stop cooling water until the column is safe to handle.
- Separate distillate and bottoms in labelled hazardous waste containers.
- Wash and put away glass. Wipe down work area.

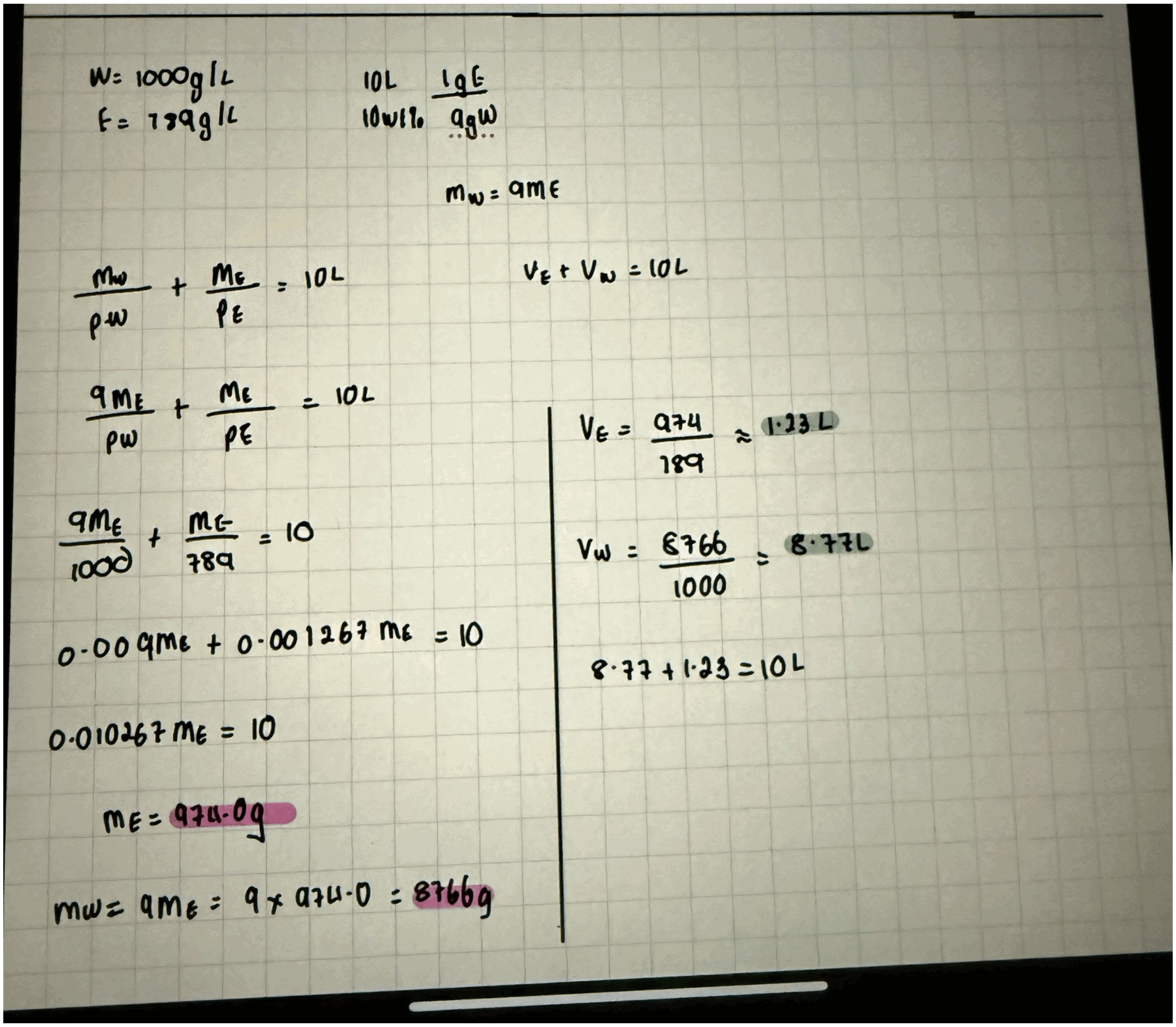
3. Project Safety Evaluation

	Yes/No	
Potential electrical hazards?	Yes	The wires can potentially be exposed to the water/steam
Potential mechanical hazards?	No	
Potential pressure hazards?	Yes	We are working with a feed pump.
Potential temperature hazards?	Yes	The distillation column will get hot from the reboiler.
Hazardous/toxic chemicals involved?	Yes	Ethanol is toxic and flammable, and must be handled in the fume hood.
Gloves required?	Yes	Heat gloves needed as ethanol solution will drip into collecting flasks at a high temperature. Gloves will also be used to measure 200 proof Ethanol inside of fume hood
MSDS reviewed by all team	Yes	Ethanol (64-17-5) Water (7732-18-5)

members?		
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	Boiling Points	Ethanol Boiling Point: 78 Water Boiling: 100 10 wt% Ethanol Mixture: 90.5
Pressure (psia)		Pressure will be measured throughout the column in cmH_2O
Safety Controls	Yes	There is a sensor in the reboiler that prevents it from turning on power if there's no feed.

4. Sample Pre-Lab Calculations

Mass balance was used to prepare 10 L of 10 wt% of ethanol solution where the densities of ethanol (0.789 g/mL) and water (1.000 g/mL) were used. It was calculated that approximately 974 g ethanol (about 1.23 L) and approximately 8766 g water (about 8.77 L) will be needed to get the final mixture.



I have read relevant background material for the Unit Operations Laboratory entitled:

"Binary Batch Distillation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

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